

Figure 11: AUROC (in percentage) of probes trained on affirmative statements and tested on negative ones. AUROC exceeding 0.5 indicates generalization success. MO refers to the Mistral-7B-v0.1 model.

BoolQ. For the BoolQ (Clark et al., 2019) benchmark, we arrange four setups, including “no options”, “with options”, “T” and “F”. We only demonstrate the prompt for “with options”. In the in-context setups, the exemplars are randomly selected from the training split.

Passage: <passage>
 Question: does ethanol take more energy
 make that produces?
 Options:
 - Yes
 - No
 Answer: No

XSum. For this task, we arrange four setups, including “zero-shot”, “T”, “TT” and “TTT”. We only demonstrate the zero-shot prompt as the one-shot and few-shot prompts can be trivially extended from it. In the in-context setups, the exemplars are randomly selected from the training split of XSum (Narayan et al., 2018) dataset, and they are all deemed correct. False examples come from the XSum Hallucination Annotations (Maynez et al., 2020) dataset with 500 examples, which is paired with examples from the test split of XSum. Furthermore, we filter for examples no longer than the LLM’s context window. For Llama-3.1-8B with a context length of 8192, we obtain the final test set of 998 examples whose labels are balanced.

Summarize this document: <doc>
 Summary: Rory McIlroy moved to within a
 shot of joint leaders Victor
 Dubuisson and Jaco van Zyl after the
 third round of the Turkish Airlines
 Open.

F.6.2 More results

We present results for Mistral-7B-v0.1 in Figure 16. Accuracy improves as in-context exemplars are provided, but calibration only displays the same trend on SciQ dataset. Another abnormality could be observed for the BoolQ dataset from “no options” setting to “with options” setting, and for

the XSum task from zero-shot to one-shot. In these cases, both accuracy and calibration worsens, which does not align with the results in Figure 7. We assume this is attributed to the weakness of the target model, where Mistral-7B-v0.1 finds it difficult to interpret answer options and in-context exemplars for the abstractive summarization task.

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The implementation of the probes is based on the scikit-learn (Pedregosa et al., 2011) library, which is licensed under BSD 3-Clause License. The factual statements we use is curated by Bürger et al. (2024), licensed under MIT License. The MMLU (Hendrycks et al., 2020) dataset is licensed under MIT License, the TriviaQA (Joshi et al., 2017) dataset is licensed under Apache 2.0 License, the SciQ (Welbl et al., 2017) dataset under Creative Commons Attribution-NonCommercial 3.0 Unported License, the BoolQ (Clark et al., 2019) under Creative Commons Share-Alike 3.0 License, the XSum (Narayan et al., 2018) dataset under MIT License and the XSum Hallucination Annotations (Maynez et al., 2020) dataset under Creative Commons Attribution 4.0 International License. Llama-2 series of models are licensed under Llama 2 Community License Agreement, Llama-3 herd of models are licensed under Llama 3 Community License Agreement and Mistral models are licensed under MIT License.

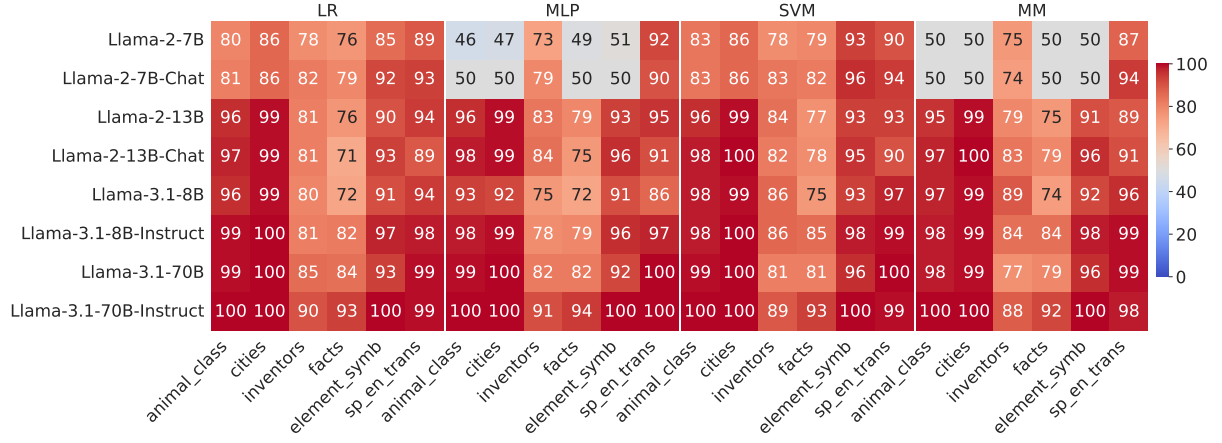


Figure 12: AUROC (in percentage) of probes trained on all the atomic factual statements and tested on logical conjunctions. AUROC > 0.5 indicates the success of generalization.

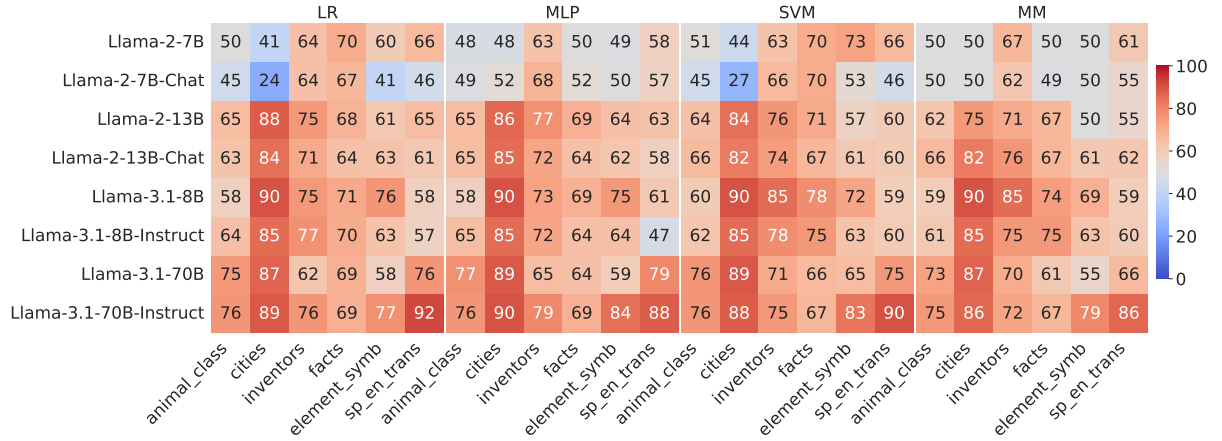


Figure 13: AUROC (in percentage) of probes trained on all the atomic factual statements and tested on logical disjunctions. AUROC > 0.5 indicates the success of generalization.

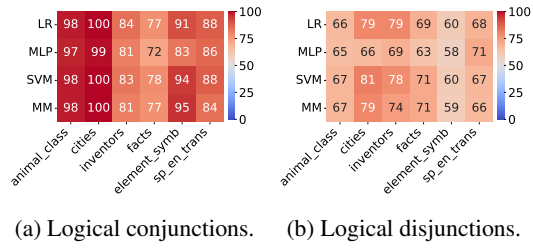


Figure 14: AUROC (in percentage) of probes trained on atomic factual statements and tested on logical conjunctions/disjunctions for Mistral-7B-v0.1. AUROC > 0.5 indicates the success of generalization.

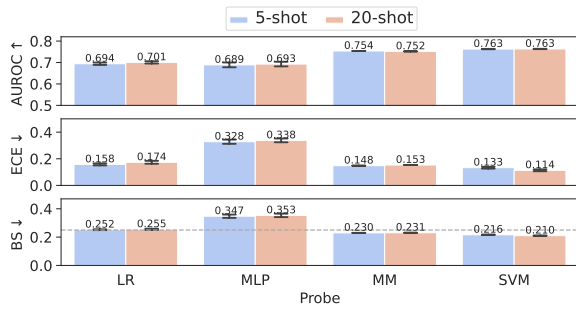
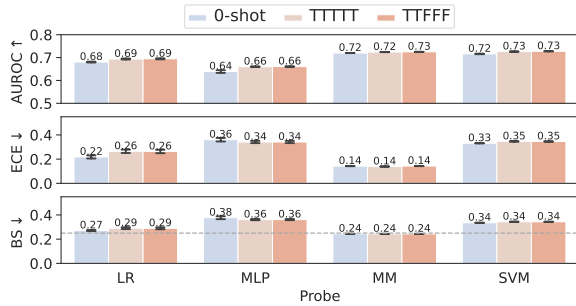


Figure 15: AUROC↑/ECE↓/BS↓ of truthfulness probes for Mistral-7B-v0.1 on MMLU and TriviaQA. The dashed gray line corresponds to random results, and error bars denote standard error.